SURYA JAL

Solar-Powered Modular Desalination and Water Purification Concept

OSFIC Problem Statement: PS-169 – Sustainable high-purity water sourcing for coastal green hydrogen and ammonia hubs

Team	Team Surya Jal
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Class	IX
Theme	Renewable Energy
Proposed application	Solar-assisted seawater desalination and a pathway to high-purity water treatment

“Turning sunlight into a sustainable water source.”

Supporting technical note | Concept proposal and prototype validation plan

1. Executive Summary

Surya Jal is a proposed solar-powered desalination concept that uses solar thermal energy to separate water from seawater through evaporation and condensation. The concept is intended to explore a renewable-energy-assisted route for producing freshwater from an abundant coastal resource.

The project is aligned with OSFIC Problem Statement PS-169, which focuses on sustainable high-purity water sourcing for coastal green hydrogen and ammonia hubs. Surya Jal is proposed as an initial desalination stage and a platform for further development—not as a currently validated industrial source of electrolyser-grade water.

Core proposition: use sunlight to drive the thermal desalination stage, measure the quantity and quality of the collected water, and investigate suitable additional purification for demanding industrial applications.

2. Problem Context and Need

Coastal industrial development can increase demand for reliable freshwater. Green hydrogen production by water electrolysis requires water that meets the equipment supplier's specified quality limits. Seawater contains dissolved salts and other constituents, so it cannot be fed directly into an electrolyser without appropriate treatment.

A sustainable water-sourcing pathway should consider the source water, energy used for desalination, final water quality, brine management, operating cost, maintenance and reliability. Surya Jal focuses on investigating solar energy as the heat source for an initial desalination step.

3. Proposed Solution and Working Principle

The proposed process has five stages:

1. Seawater feed: Saline water is introduced into an enclosed evaporation chamber.

2. Solar absorption and evaporation: A dark absorber captures solar radiation and transfers heat to the saline water. Water evaporates while most dissolved salts remain in the chamber.

3. Condensation: Water vapour contacts a cooler, inclined transparent cover and condenses into droplets.

4. Collection: The droplets flow into a separate channel or container, avoiding contact with the remaining brine.

5. Quality testing and optional polishing: The collected water is tested. Additional treatment may be required depending on the intended use.

4. Main Components and Their Functions

Component	Proposed function	Design consideration
Dark solar absorber	Absorbs sunlight and converts it into heat.	Compare coating/material options; check corrosion resistance and safe contact with water.
Evaporation chamber	Holds saline feed and contains warm vapour.	Insulation, leak prevention, cleanability and safe operation.
Transparent inclined cover	Provides a condensation surface and guides droplets.	Slope, surface cleanliness, cooling and secure fitting.
Collection channel	Separates and collects condensed water.	Prevent mixing with feedwater or brine; use clean collection vessels.
Testing instruments	Measure temperature, volume and water quality indicators.	Use calibrated meters where possible; record units and test conditions.

5. Proposed Innovation and Development Features

The initial prototype demonstrates the basic evaporation–condensation principle. The following features are proposed for investigation and should be described as development goals until they are tested:

• **Modular design:** explore repeatable units that can be connected to increase capacity.

• **Thermal optimisation:** compare absorber surfaces, insulation and chamber geometry to reduce heat loss.

• **Improved condensation:** investigate cover angle, surface temperature and collection-channel design.

• **Heat recovery:** assess whether residual heat can pre-warm incoming saline feed.

• **Hybrid purification:** study the need for reverse osmosis, ion exchange or deionisation after the thermal stage.

• **Performance monitoring:** use temperature, output-volume and conductivity/TDS measurements to guide design changes.

6. Relevance to Green Hydrogen and Ammonia Hubs

Surya Jal's potential role is as a solar-assisted pre-treatment or desalination concept within a larger water-treatment train. A realistic industrial pathway would require seawater intake and pre-treatment, desalination, appropriate polishing (such as reverse osmosis and/or deionisation), and final quality assurance against the electrolyser manufacturer's specification.

Important limitation: solar distillation can remove most non-volatile dissolved salts from the condensed water, but it does not by itself prove that the product meets high-purity industrial requirements. Water quality must be measured, and additional treatment may be necessary. No claim of electrolyser-grade water should be made without validated test results.

7. Prototype Testing and Validation Plan

The prototype should be evaluated through repeatable tests. Record the date, test duration, sunlight/weather conditions, feedwater volume, temperatures, collected-water volume and water-quality readings. Repeat tests where possible so that results are not based on a single observation.

Parameter	Method / record	Result
Feedwater volume	Measure starting seawater/saline-water volume (mL or L).	To be measured
Collected water volume	Measure condensate collected after a fixed test period.	To be measured
Temperature	Record absorber, feedwater and ambient temperature (°C), if instruments are available.	To be measured
Conductivity or TDS	Measure feedwater and collected water using a suitable meter; note units.	To be measured
Operating conditions	Record start/end time, cloud cover, wind and any interruptions.	To be recorded
Visible observations	Note leakage, condensation, scaling, corrosion or contamination.	To be recorded

Suggested calculations

Water recovery (%) = (collected water volume ÷ feedwater volume) × 100. This is a volume recovery indicator, not a complete measure of energy efficiency or salt rejection.

Change in TDS or conductivity should be calculated only from paired measurements taken with the same instrument and units. TDS meters provide an estimate based on conductivity; they do not identify every contaminant.

8. Safety, Quality and Environmental Considerations

• Use the prototype under adult/teacher supervision, especially when working with hot surfaces, glass, electrical instruments or tools.

• Do not seal a heated vessel in a way that can create pressure. The system must remain safely vented as designed.

• Use stable supports and handle glass or sharp-edged materials carefully.

• Keep the collected-water container clean and covered. Do not drink prototype water or describe it as potable without appropriate testing and approval.

• Plan for the concentrated brine left after evaporation. Avoid uncontrolled discharge into soil, drains or freshwater environments.

• Choose corrosion-resistant, cleanable materials suitable for saline conditions.

9. Limitations and Risk Reduction

Limitation / risk	Why it matters	Possible response
Weather dependence	Solar heat varies with cloud cover and season.	Test across different conditions; consider thermal storage or hybrid heat input in future research.
Low output from a small still	A simple solar still may produce limited water per unit area.	Measure actual output; investigate larger area, multiple stages and improved condensation.
High-purity requirement	Condensate may still contain volatile contaminants or become contaminated during collection.	Test quality and add appropriate polishing; follow end-use specifications.
Scaling and corrosion	Salts can deposit and seawater can corrode materials.	Select suitable materials; clean and inspect regularly.
Brine management	Concentrated brine needs responsible handling.	Collect and dispose of brine responsibly; assess environmental impact before scaling.

10. Development Roadmap

Phase	Focus	Expected output
1	Build and baseline test	Working proof-of-concept and initial measurements.
2	Design optimisation	Comparison of absorber, insulation, cover angle and operating conditions.
3	Water-quality assessment	Documented conductivity/TDS results and assessment of additional purification.
4	Scale and feasibility study	Modular design proposal with output, cost, maintenance and environmental assessment.

11. Conclusion

Surya Jal proposes a practical starting point for exploring solar-powered seawater desalination. Its value lies in combining a demonstrable scientific principle with a clear research pathway: improve thermal performance, measure freshwater output and quality, and assess how additional purification could make the process relevant to demanding coastal applications.

The project is a concept under development. Its industrial relevance will depend on measured performance, water-quality validation, cost analysis, safe brine management and further engineering.

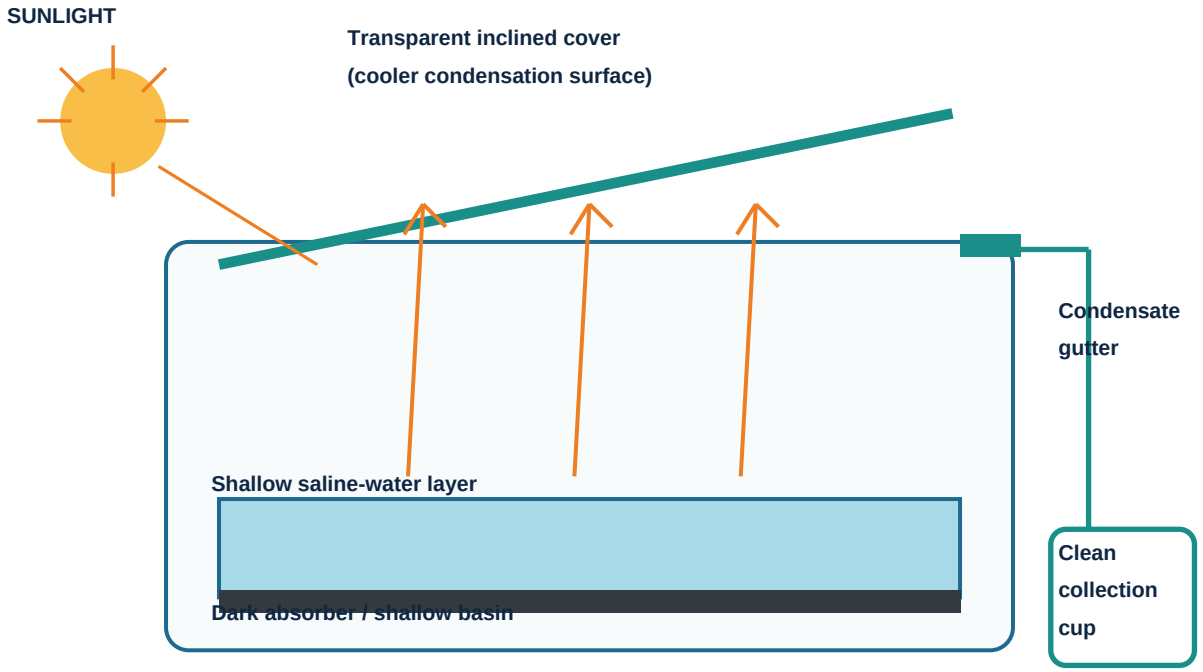
12. References and Further Reading

• International Renewable Energy Agency (IRENA) – resources on renewable energy and green hydrogen: https://www.irena.org/
• U.S. Department of Energy – Hydrogen and Fuel Cell Technologies Office: https://www.energy.gov/eere/fuelcells/hydrogen-and-fuel-cell-technologies-office
• World Health Organization – Guidelines for Drinking-water Quality (for general water-quality context; industrial electrolyser specifications are different): https://www.who.int/publications/i/item/9789240045064
• Electrolyser manufacturer's technical documentation should be consulted for the exact feedwater quality requirements of any proposed industrial application.

Submission note: Add photographs of the actual prototype and replace "To be measured" fields with genuine observations before uploading. Do not present planned features or unmeasured performance as achieved results.

13. Recommended Working Model: Single-Slope Solar Still

Recommended prototype: a single-basin, single-slope solar still with a dark stainless-steel absorber, shallow saline-water tray, insulated enclosure, inclined transparent cover and separate condensate gutter. This is the most practical base model to demonstrate the Surya Jal principle reliably at student-prototype scale.



Conceptual single-slope solar still — not to scale. Keep the chamber safely vented; never build a pressure vessel.

How to read the diagram: sunlight heats the dark basin; water evaporates from the shallow saline layer; vapour contacts the cooler inclined cover; droplets run down the slope into a gutter and then a separate clean collection cup.

13.1 Suggested prototype layout

Part	Suggested student-prototype specification	Purpose
Base / enclosure	Rigid, insulated box or tray; approx. 45 × 30 cm footprint as a starting layout, adjusted to available materials.	Supports components and reduces heat loss.
Evaporation basin	Shallow stainless-steel tray or basin that fits inside the enclosure.	Holds saline feed and absorbs heat.
Solar absorber	Dark, heat-absorbing surface beneath/at the basin; use a suitable non-toxic, heat-safe coating only on the outside if needed.	Improves solar heat absorption.
Transparent cover	Glass or suitable clear, heat-resistant transparent sheet mounted at a slope; secure edges without making a sealed pressure chamber.	Provides a condensation surface and directs droplets.
Gutter + outlet	Small clean channel fixed at the lower edge of the cover, connected to a clean collection cup.	Collects condensate separately.
Insulation	Insulating layer around the outside of the chamber, kept away from direct flame or excessive heat.	Reduces unwanted heat loss.

13.2 Assembly sequence

• Place the shallow stainless-steel basin inside the insulated base. Ensure it is stable and easy to remove for cleaning.

• Position the dark absorber so that it receives sunlight and transfers heat to the basin. Avoid coatings that could contact the water.

• Add a shallow layer of saline test water to the basin. A shallow layer heats more readily than a deep tank; record the exact starting volume.

• Fit the transparent cover at an incline, with its lower edge positioned above the collection gutter. The inside surface should be clean.

• Attach the gutter to a separate collection cup. Check that droplets can flow without leaking back into the saline basin.

• Keep the enclosure safely vented according to the design. Do not seal it airtight or allow pressure to build up.

• Place the model in direct sunlight on a stable, level support. Monitor it under adult/teacher supervision.

14. How to Demonstrate and Evaluate the Model

14.1 Demonstration steps

• Show the starting saline water and explain that it is the feedwater.

• Point to the dark absorber and explain how it converts solar radiation into heat.

• Explain that water vapour rises while most dissolved salts remain in the basin.

• Show droplets forming on the underside of the inclined cover and moving towards the gutter.

• Collect the condensate in a separate clean container and measure its volume after a fixed test period.

• Compare feedwater and collected-water conductivity/TDS using a suitable meter, recording the instrument and units.

14.2 Fair-test checklist

Keep constant	Change / measure	Record
Basin area, starting volume, test duration and cover arrangement for comparison tests.	One design factor at a time (for example, absorber surface or cover slope).	Date, sunlight/cloud cover, ambient conditions, temperatures and collected volume.
Use the same measuring method and instrument where possible.	Repeat each condition on more than one occasion if possible.	Feedwater and condensate TDS/conductivity, leakage, visible scaling and observations.

14.3 Suggested improvement sequence

First: establish a reliable baseline with the simple single-slope design. **Second:** compare insulation and absorber options. **Third:** improve the cover angle and gutter arrangement. **Later research:** investigate heat recovery or multi-effect stages. Do not add complex features until the basic model produces repeatable measurements.

15. Technical Limits and Honest Claims

This student-scale model demonstrates solar evaporation and condensation; it is not an industrial desalination plant. Output is strongly affected by sunlight, exposed area, insulation, cover temperature and weather. A small solar still may produce limited water, so its practical value must be judged from measured output and cost per litre.

The collected condensate must not be assumed safe to drink or suitable for electrolysis. Test water quality and use further treatment when required. Industrial use would need engineering design, validated performance, monitoring, maintenance planning and responsible brine management.

16. Photograph placeholders for your final submission

Insert your own photographs in the spaces below before uploading: (1) full working model, (2) close-up of the inclined cover and collection gutter, (3) solar absorber and saline basin, and (4) water-quality testing or measurement setup.

Photo 1: Complete model
Photo 2: Condensation cover + gutter
Photo 3: Absorber + basin

Photo 4: Testing / measurements